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The effect of prolonged slump sitting on postural sway in female athletes and non-athletes

Reihaneh Zarnegar¹ , Hooman Minoonejad^{1*} , Mohammad Karimizadeh Ardakani¹ and Ebrahim Ebrahimi¹

Abstract

Purpose Prolonged sedentary behavior and slump sitting posture may impair neuromuscular function; however, acute effects on postural sway and potential differences between athletes and non-athletes remain unclear. Therefore, this study aims to examine the acute effects of prolonged slump sitting on postural sway in female athletes and non-athletes.

Methods In this study, 24 females (12 athletes, 12 non-athletes; age 18–30 years) completed balance assessments using the Biodex Balance System, followed by a 30-minute standardized slump sitting protocol. Outcomes included static postural stability test (PST), Modified Clinical Test of Sensory Interaction on Balance (mCTSIB), and limits of stability (LOS). Group differences were analyzed using analysis of covariance (ANCOVA).

Results Prolonged slump sitting significantly influenced selected postural sway outcomes. Athletes demonstrated greater post-intervention sway compared with non-athletes in the Overall Stability Index ($p=0.01$) and the eyes-closed firm-surface mCTSIB condition ($p=0.02$). A significant group difference was also observed in static left LOS performance ($p=0.01$).

Conclusion In a nutshell, the present study demonstrated that prolonged slump sitting may influence postural sway in female athletes and non-athletes, with specific differences emerging between groups under certain sensory and stability conditions. By highlighting the potential impact of slump sitting on postural stability, this research contributes to the fields of sports science, ergonomics, and rehabilitation, emphasizing the need for strategies that mitigate the negative effects of prolonged sedentary behavior.

Keywords Posture, College Athletes, Postural Sway

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Introduction

Sedentary behavior has emerged as an essential public health concern, with prolonged sitting increasingly recognized as a contributor to musculoskeletal dysfunction [1], impaired neuromuscular control, and decreased postural stability [2]. Modern occupational and academic environments often require extended periods of static sitting, frequently in suboptimal postures such as slump sitting, a posture characterized by posterior pelvic tilt, thoracolumbar flexion, and forward head positioning [3]. Prolonged slump sitting has been shown to induce muscle fatigue [4], reduce lumbar lordosis, alter spinal loading patterns [5], and decrease activation of deep stabilizing muscles, including the transversus abdominis and multifidus [6]. Furthermore, sustained flexed postures may influence sensorimotor integration through changes in mechanoreceptor sensitivity and neuromuscular recruitment patterns [7, 8]. These biomechanical changes may impair proprioceptive feedback from spinal and pelvic structures [9], disrupting postural control strategies during subsequent standing tasks [10].

Postural control is a complex process involving the integration of sensory input from visual, vestibular, and proprioceptive systems with motor responses to maintain balance during static and dynamic tasks [11]. A commonly used quantitative indicator of postural control is postural sway, typically assessed through center-of-pressure (COP) displacement, velocity, and variability during quiet standing [12]. Increased postural sway has been associated with impaired neuromuscular coordination [13], fatigue [14], reduced proprioceptive acuity [15], and elevated injury risk. In sports science, postural sway metrics have been widely used to evaluate injury risk, rehabilitation progress, and training adaptations [16], with evidence suggesting that athletes often exhibit more efficient balance strategies compared to non-athletes [17, 18]. In athletic populations, optimal postural control is crucial not only for performance but also for injury prevention [19], as deficits in balance and neuromuscular stability have been linked to conditions such as anterior cruciate ligament (ACL) injuries [20], ankle sprains [21], and chronic low back pain [22]. A study showed that individuals who underwent ACL reconstruction appeared to exhibit a persistent postural balance deficit, particularly in dynamic double-leg conditions [23].

Despite increasing research on sedentary behavior and postural control, important gaps remain. Existing studies have largely emphasized chronic musculoskeletal outcomes of sitting rather than the effects of prolonged slump sitting on postural sway [24]. Direct comparisons between athletes and non-athletes following standardized sedentary postures are limited, leaving unclear whether athletic conditioning mitigates or amplifies transient balance disturbances. Female-specific investigations

are also underrepresented despite known sex-related differences in neuromuscular control [25], balance strategies, and injury susceptibility. Furthermore, variability in sitting protocols, posture standardization, and sway assessment methods has restricted the development of clear recommendations that integrate ergonomic, sports performance, and rehabilitation perspectives. Therefore, the present study aimed to investigate the effects of prolonged slump sitting on postural sway in female athletes and non-athletes. It was further hypothesized that the effects of prolonged slump sitting on postural sway would be condition-specific and more pronounced under sensory-challenged conditions.

Materials and methods

The present study employed a causal-comparative with pre-test and post-test design. Participants were recruited using convenience sampling from the Faculty of Sport Sciences and Health at the University of Tehran. Based on a study [26], 24 participants were chosen using G. Power software, ensuring a 0.80 test power, an effect size of 0.84, and a significance level of 0.05. They were divided into athletes ($N = 12$) and non-athletes' groups ($N = 12$). Participants were eligible for inclusion if they met the following criteria: female sex; age between 18 and 30 years; body mass index (BMI) between 18.5 and 25 kg/m²; no history of fracture or surgical intervention involving the shoulder, lumbar spine, or lower extremities within the previous six months; absence of neurological disorders, stroke, or functional dysfunction in the neck or shoulder regions; and ability to safely perform all testing procedures. Also, the athlete group consisted of individuals with a minimum of three years of regular participation in organized sports activities, defined as structured and supervised training programs conducted within university teams or sports clubs. Participants were engaged in various disciplines, including team sports (e.g., volleyball, basketball) and individual sports (e.g., athletics). All athletes trained at least three sessions per week, with each session lasting a minimum of 90 min. The majority of participants competed at a recreational to university level rather than elite or professional levels. Participants were excluded if they had engaged in intense physical activity within the 24 h preceding testing, were outside the specified age range, demonstrated non-compliance with study procedures, or chose to withdraw from participation at any point during the study. Based on the Declaration of Helsinki, all stages of the study were explained to the participants, and then written informed consent was obtained. Secondly, individuals were informed that in the event of any issues during the tests, the examiner, a sports science student, would take all necessary actions. Also, the participants were instructed on how to perform the tests. Demographic and anthropometric

characteristics of the participants, including age, height, and body mass, were recorded using a standardized demographic proforma completed prior to testing. Body height was measured using a wall-mounted stadiometer to the nearest 0.1 cm, and body mass was measured using a calibrated digital scale to the nearest 0.1 kg, with participants barefoot and wearing light clothing. BMI was calculated as body mass in kilograms divided by height in meters squared (kg/m^2). All steps were explained to the participants. It is worth mentioning that all tests were conducted in the laboratory of Tehran University, Iran. Before starting the investigation, study approval was obtained from the Biomedical Research Ethics Committee of Tehran University (Ethics code: IR.UT.SPORT.REC.1403.08).

Procedure

The test procedure was conducted as follows: initially, a pre-test assessment of postural sway using the Biodex Balance System was performed for both groups. In the next stage, participants were positioned in a slump sitting posture (seated on a backless chair in a flexed position, with the hip and knee joints at 90° flexion, and a

posture in which the angle formed between the acromion, L1 vertebra, and greater trochanter deviated by $\pm 3^\circ$ from 93°) [4]. This angle was established using anatomical landmarks and measured with a handheld goniometer, supported by visual alignment (Fig. 1). Prior to the intervention, the examiner adjusted each participant's posture to meet the required criteria. Participants maintained the slump sitting position for 30 min [4], after which post-test measurements of postural sway were recorded. During this period, posture maintenance was continuously monitored by the examiner. Participants were instructed to maintain the defined slump position and avoid postural corrections. Verbal reminders were provided when necessary to ensure adherence, and any substantial deviations from the target posture were corrected immediately.

Postural sway assessment

Postural sway was assessed using the Biodex Balance System. All participants completed the Postural Stability Test (PST), Modified Clinical Test of Sensory Interaction on Balance (mCTSIB), and Limits of Stability (LOS) test according to standardized manufacturer guidelines under the supervision of an experienced examiner. Reported reliability coefficients demonstrated the Biodex Balance System as a valid and reliable tool for assessing postural sway ($R = 0.94$) [27]. Prior to testing, participants received a detailed explanation of procedures and performed familiarization trials to minimize learning effects. For all assessments, participants stood barefoot on the Biodex platform in a standardized upright posture with feet positioned according to the device's coordinate grid. Foot placement was recorded during the initial trial and replicated in subsequent measurements to ensure consistency between pre-test and post-test sessions. Participants were instructed to maintain an upright standing posture with arms alongside the body unless otherwise specified and to focus on the visual feedback displayed on the monitor [28].

Postural Stability Test (PST)

To evaluate static postural control, participants completed the PST [28]. They were instructed to stand in a double-leg stance with eyes open (DLEO) and maintain the platform indicator at the center of the display while minimizing body sway. At the beginning of the test, the participant's foot position and stance width were recorded. During each trial, once the device was activated, participants were required to keep the platform as stable and horizontal as possible for 15 s, preventing excessive deviations and disturbances. The stability level of the movable platform was adjustable; in this study, it was set between levels 3 and 6 during double-leg stance trials to provide moderate instability while maintaining



Fig. 1 Slump sitting posture based on the given definition

Table 1 Demographic characteristics of participants

Data	Group	Mean $\pm$ SD	P-value
Age (year)	Athletes	24.81 $\pm$ 3.60	0.58
	Non-athletes	25.66 $\pm$ 3.16	
Height (cm)	Athletes	163.09 $\pm$ 5.88	0.05
	Non-athletes	167.88 $\pm$ 4.28	
Weight (kg)	Athletes	59.63 $\pm$ 6.07	0.20
	Non-athletes	60.77 $\pm$ 6.11	
BMI (kg/m ²)	Athletes	23.43 $\pm$ 2.35	0.96
	Non-athletes	24.55 $\pm$ 3.45	

BMI Body Mass Index

Table 2 ANCOVA results of static PST between athletes and non-athletes

Variable	Group	Pre-Mean $\pm$ SD	Post-Mean $\pm$ SD	F	Eta squared	P-value
OSI	Athletes	0.90 $\pm$ 0.32	0.64 $\pm$ 0.13	7.74	0.31	0.01*
	Non-athletes	0.31 $\pm$ 0.14	0.52 $\pm$ 0.10			
APSI	Athletes	0.28 $\pm$ 0.14	0.45 $\pm$ 0.17	0.71	0.03	0.41
	Non-athletes	0.23 $\pm$ 0.08	0.45 $\pm$ 0.14			
MLSI	Athletes	0.16 $\pm$ 0.06	0.33 $\pm$ 0.16	0.25	0.01	0.62
	Non-athletes	0.15 $\pm$ 0.08	0.30 $\pm$ 0.11			

OSI Overall Stability Index, APSI Anterior–Posterior Stability Index, MLSI Medial–Lateral Stability Index

participant safety. The Biodex system automatically calculated and recorded three primary stability indices: Overall Stability Index (OSI), Anterior–Posterior Stability Index (APSI), and Medial–Lateral Stability Index (MLSI).

Modified Clinical Test of Sensory Interaction on Balance (mCTSIB)

The mCTSIB was conducted to assess the contribution of visual and somatosensory inputs to postural control under different sensory conditions. Each trial required participants to stand quietly while maintaining stability on the platform for the specified test duration according to Biodex protocols. Participants were instructed to remain as still as possible and avoid excessive movements [28]. The device recorded sway indices and postural stability performance across sensory conditions, allowing evaluation of sensory integration and balance control mechanisms. Participants completed four standardized conditions: Eyes open with a firm surface (EO-firm), Eyes closed with a firm surface (EC-firm), Eyes open with a foam surface (EO-foam), and Eyes closed with a foam surface (EC-foam).

Limits of Stability (LOS) Test

Dynamic balance and voluntary control of the center of gravity (COG) were evaluated using the LOS test. Participants stood on the platform at stability level 5 with feet positioned according to recorded coordinates and arms crossed over the chest. While viewing the monitor, participants were instructed to shift their body weight and move their COG toward randomly appearing visual

targets in multiple directions (anterior, posterior, left, and right). Targets appeared sequentially and unpredictably, requiring controlled and accurate weight shifting without lifting the feet [28]. Participants were asked to move their COG toward each target as quickly and accurately as possible. The Biodex system calculated parameters related to movement time, directional control, and dynamic stability performance. All balance assessments were performed before and immediately after the prolonged slump sitting intervention under identical testing conditions to evaluate changes in postural sway and stability performance.

Statistical analysis

Data analysis was performed using SPSS software (version 27). The normality of the data distribution was first examined using the Shapiro–Wilk test. Differences between athletes and non-athletes, while controlling for pre-test scores (covariate), were analyzed using analysis of covariance (ANCOVA). For each comparison, F-values, eta squared (η^2) as a measure of effect size, and p-values were calculated. Statistical significance was set at $p \leq 0.05$.

Results

Table 1 shows the demographic characteristics of participants in both groups. There were no significant differences in demographic data among the study groups.

The results of Table 2 indicated that there was a significant difference between the athlete and non-athlete groups in the OSI ($p=0.01$, eta squared=0.31), even when the pre-test effect was taken into account

Table 3 ANCOVA results of mCTSIB between athletes and non-athletes

Variable	Group	Pre-Mean $\pm$ SD	Post-Mean $\pm$ SD	F	Eta squared	P-value
EO-firm	Athletes	1.03 $\pm$ 0.50	1.47 $\pm$ 0.80	0.01	0.01	0.92
	Non-athletes	0.96 $\pm$ 0.65	1.53 $\pm$ 1.27			
EC-firm	Athletes	0.28 $\pm$ 0.14	0.45 $\pm$ 0.17	0.92	0.25	0.02*
	Non-athletes	0.32 $\pm$ 0.08	0.45 $\pm$ 0.14			
EO-foam	Athletes	49.09 $\pm$ 14.25	70.09 $\pm$ 9.74	1.82	0.09	0.19
	Non-athletes	44.22 $\pm$ 12.29	67.88 $\pm$ 11.45			
EC-foam	Athletes	56.72 $\pm$ 18.10	72.63 $\pm$ 14.05	0.29	0.01	0.59
	Non-athletes	47.66 $\pm$ 18.86	68.55 $\pm$ 18.52			

Table 4 ANCOVA results of the LOS test between athletes and non-athletes

Variable	Group	Pre-Mean $\pm$ SD	Post-Mean $\pm$ SD	F	Eta squared	P-value
Anterior LOS- static	Athletes	56.72 $\pm$ 18.10	72.63 $\pm$ 14.05	0.53	0.03	0.47
	Non-athletes	47.66 $\pm$ 18.86	68.55 $\pm$ 18.52			
Posterior LOS- static	Athletes	53.45 $\pm$ 15.68	76 $\pm$ 11.31	0.92	0.05	0.34
	Non-athletes	50.55 $\pm$ 14.62	72.22 $\pm$ 7.20			
Right LOS- static	Athletes	55 $\pm$ 17.52	70.36 $\pm$ 12.20	0.71	0.03	0.41
	Non-athletes	51.55 $\pm$ 12.28	69.55 $\pm$ 10.26			
Left LOS- static	Athletes	58.90 $\pm$ 12.87	72.27 $\pm$ 12.30	7.74	0.31	0.01*
	Non-athletes	49.77 $\pm$ 12.26	71.22 $\pm$ 7.77			
Anterior LOS- dynamic	Athletes	56.72 $\pm$ 18.10	72.63 $\pm$ 14.05	0.43	0.02	0.52
	Non-athletes	43.11 $\pm$ 16.88	44 $\pm$ 12.89			
Posterior LOS- dynamic	Athletes	62.54 $\pm$ 12.69	61 $\pm$ 13.06	1.23	0.06	0.28
	Non-athletes	57.44 $\pm$ 16.07	49.77 $\pm$ 22.21			
Right LOS- dynamic	Athletes	49.27 $\pm$ 14.86	55.63 $\pm$ 15.53	0.18	0.01	0.67
	Non-athletes	43.44 $\pm$ 20.67	48.77 $\pm$ 17.04			
Left LOS- dynamic	Athletes	52.09 $\pm$ 15.53	58.90 $\pm$ 14.25	0.34	0.02	0.56
	Non-athletes	42.33 $\pm$ 18.10	50.88 $\pm$ 17.94			

LOS Limits of Stability

(covariate), indicating higher scores in the athlete group compared to the non-athlete group.

EO-firm Eyes open with a firm surface, EC-firm Eyes closed with a firm surface, EO-foam Eyes open with a foam surface, EC-foam Eyes closed with a foam surface.

The results of Table 3 showed that there was a significant difference between the athlete and non-athlete groups in the EC-firm ($p=0.02$, eta squared=0.25), even when the pre-test effect was taken into account (covariate), demonstrating higher scores in the athlete group compared to the non-athlete group. The reported Mean $\pm$ SD values represent the sway index, derived from COP displacement, under each sensory condition.

The results of Table 4 revealed that there were no significant differences between the athlete and non-athlete groups in all dynamic and static LOS tests ($P \geq 0.05$), even when the pre-test effect was taken into account (covariate), except for the static left LOS test ($p=0.01$, eta squared=0.31). The Mean $\pm$ SD values in this table represent directional control (%).

Discussion

The purpose of the present study was to investigate the effects of prolonged slump sitting on postural sway in female athletes and non-athletes. The main findings indicated that prolonged slump sitting influenced several postural stability outcomes, with significant differences observed between groups in OSI, the EC-firm condition of the mCTSIB, and the static left LOS test. Athletes demonstrated higher post-intervention sway scores in selected conditions compared to non-athletes, suggesting that prolonged slump sitting may temporarily alter postural control mechanisms regardless of athletic status. However, most dynamic LOS variables and other mCTSIB conditions did not show significant group differences, indicating that the effects may be task-specific and more evident under certain sensory or stability demands.

In respect of OSI, the observed increase in postural sway is consistent with studies showing that prolonged slumped or flexed sitting postures may contribute to neuromuscular fatigue and altered postural control. Jung et al. (2021) reported increased trunk muscle fatigue after

prolonged slumped sitting [4], which may negatively influence postural stability during subsequent standing tasks. Similarly, de Groot et al. (2014) found that flexed posture is associated with impaired postural control, supporting the notion that sustained spinal flexion affects balance regulation [10]. However, these findings contrast with research suggesting that athletes generally exhibit lower postural sway and more efficient stability strategies than non-athletes. Donath et al. (2016) observed improved balance performance in trained individuals during challenging postural tasks [13], while Andreeva et al. (2021) reported superior postural stability among athletes [14]. The discrepancy may be attributed to the acute fatigue and proprioceptive alterations induced by prolonged slump sitting, which may temporarily reduce the neuromuscular advantages typically observed in athletic populations. Several mechanisms may explain the increase in OSI observed after prolonged slump sitting. Sustained spinal flexion may reduce lumbar lordosis and alter muscle activation patterns [29], particularly decreasing the engagement of deep stabilizing muscles such as the transversus abdominis and multifidus [30]. This may reduce trunk stiffness and impair the ability to control COP movements during standing tasks [31]. Additionally, prolonged static posture may disrupt proprioceptive feedback [9], leading to decreased sensorimotor integration. The higher OSI values observed in athletes may reflect a temporary disruption of finely tuned motor control systems that rely on precise neuromuscular coordination. Athletes often employ adaptable and dynamic balance strategies [32], and acute fatigue or altered sensory input may result in increased sway variability rather than rigid stabilization. Within the broader field, these findings highlight that acute sedentary behavior may transiently affect postural control even in physically trained individuals.

Concerning the results of mCTSIB, athletes demonstrated higher post-test sway scores compared to non-athletes, indicating greater postural instability when visual input was removed. A study showed that prolonged slumped sitting may compromise proprioceptive input and postural control when visual cues are unavailable [33]. Similarly, Peterka (2002) highlighted the importance of sensorimotor integration for maintaining balance [8], suggesting that disruptions in proprioceptive feedback can impair stability [34], particularly under eyes-closed conditions. Conversely, previous literature has often shown that athletes possess superior sensory integration and balance performance compared to non-athletes. For example, Gaerlan et al. (2012) reported enhanced postural control among trained individuals under sensory-challenged conditions [11], and Donath et al. (2016) found that athletes generally demonstrate reduced sway during demanding balance tasks [13]. The contrasting

results in the present study may reflect the acute neuromuscular and sensory alterations caused by prolonged slump sitting, which may disrupt the sensory weighting strategies typically advantageous in athletes. The significant changes observed in the EC-firm condition likely reflect a disruption in proprioceptive processing and central sensory reweighting mechanisms following prolonged slump sitting [35]. When visual input is removed, the postural control system must rely primarily on somatosensory and vestibular information to maintain stability. Sustained spinal flexion may alter afferent input from paraspinal muscles [36], ligaments, and joint mechanoreceptors [37], thereby degrading the quality of proprioceptive signals reaching the central nervous system (CNS). Prolonged slump sitting may also induce neuromuscular fatigue in deep trunk stabilizers and alter feed-forward activation patterns necessary for anticipatory postural adjustments [38]. Reduced activation or delayed recruitment of stabilizing muscles may impair the ability to maintain equilibrium when sensory demands increase [39]. Furthermore, when vision is removed, individuals may demonstrate increased sway due to insufficient proprioceptive compensation. The greater sway observed in athletes may be interpreted through the aspect of motor control flexibility and sensory dependence. Athletes often develop highly refined sensorimotor strategies optimized for dynamic movement and rapid environmental interaction [40]. However, these systems may be more sensitive to acute perturbations such as fatigue or altered proprioceptive input, resulting in temporary instability under sensory-deprived conditions. Another possible explanation is that athletes utilize more exploratory postural strategies [41], reflecting adaptive motor variability rather than purely reduced stability.

Additionally, a significant difference between female athletes and non-athletes was observed only in the static left LOS parameter after the intervention, suggesting increased sway during voluntary weight shifting toward the left side. The observed changes align with literature indicating that prolonged flexed postures can disrupt trunk stability and alter directional weight-shifting performance. O'Sullivan et al. (2006) demonstrated that variations in sitting posture may influence spinal-pelvic alignment and trunk muscle activation patterns [5], potentially affecting postural control during subsequent standing movements. Similarly, Windhorst (2007) emphasized the importance of proprioceptive feedback from spinal musculature in guiding directional balance responses [9], suggesting that altered afferent input could impair targeted stability tasks such as LOS. However, previous research has often shown that athletes possess superior directional control and LOS compared to non-athletes. Parker et al. (2008) reported more efficient balance and motor control strategies in athletic

populations [42], and Andreeva et al. (2021) found that trained individuals generally demonstrate enhanced postural adaptability across multiple directions [14]. The differences observed in the present study may reflect the acute neuromuscular effects of prolonged slump sitting, which may override typical athletic advantages by disrupting trunk control and sensorimotor coordination. The direction-specific impairment in the static left LOS may reflect complex interactions between biomechanical alignment, neuromuscular fatigue, and hemispheric motor control mechanisms following prolonged slump sitting [43]. Sustained spinal flexion may induce asymmetric loading patterns on the pelvis and lumbar spine [44], potentially leading to uneven activation or fatigue of lateral trunk stabilizers such as the quadratus lumborum and oblique abdominal muscles [45]. Because LOS tasks require controlled displacement of the COG toward the boundaries of stability, even subtle asymmetries in muscle activation or proprioceptive signaling may influence performance in a single direction. In addition, slump sitting may alter intersegmental coordination between the trunk and lower extremities [46], which could reduce the efficiency of anticipatory postural adjustments necessary for precise lateral weight shifts. From a neurophysiological perspective, prolonged static posture may disrupt cortical and subcortical motor planning processes involved in directional balance control [47]. It is also possible that lateralized control of movement, influenced by limb dominance and hemispheric specialization, may make one direction, such as the left, more vulnerable to fatigue-induced motor variability [48]; however, these mechanisms were not directly assessed in the present study and should be interpreted with caution. The greater changes observed in athletes may reflect their reliance on high-gain neuromuscular control strategies that are optimized for performance but sensitive to perturbations [49]. When proprioceptive input is altered or stabilizing musculature is fatigued, athletes may exhibit increased movement amplitude or exploratory sway as their motor system attempts to recalibrate balance boundaries [50]. Within the broader framework of postural control, these findings suggest that acute postural stressors can impair directional stability [51], emphasizing the importance of symmetrical trunk function and sensorimotor integrity for maintaining optimal limits of stability.

The existing literature presents inconsistent findings regarding postural control differences between athletes and non-athletes, which may be explained by several moderating factors. A recent review highlighted that although athletes generally demonstrate superior balance performance compared to non-athletes, these differences are highly dependent on sport-specific demands, level of expertise, and testing conditions [52]. For example, athletes engaged in disciplines requiring high levels of static or dynamic

balance (e.g., gymnastics, shooting, or martial arts) tend to exhibit reduced postural sway, whereas athletes from other sports may not demonstrate the same advantage [52]. Moreover, the influence of visual input is critical; differences between athletes and non-athletes often become more pronounced under eyes-closed conditions, where reliance on proprioceptive and vestibular systems increases [53, 54]. Furthermore, fatigue and acute physiological stressors can temporarily impair postural stability, potentially diminishing the typical balance advantages observed in athletes [55]. In addition, training experience and competitive level appear to play a critical role in postural control performance. Higher-level athletes generally exhibit more refined neuromuscular control and reduced sway compared to lower-level or recreational athletes [56]; however, this relationship is not always consistent across studies and may depend on the interaction between sensory conditions and task complexity [52].

The findings have practical implications for athletes, students, and individuals engaged in prolonged sedentary activities. From a clinical perspective, minimizing sustained slump sitting and incorporating regular movement breaks may help maintain optimal neuromuscular function and postural stability. For sports practitioners and coaches, awareness of the potential transient balance disturbances following prolonged sitting may inform warm-up strategies and injury prevention programs. Theoretically, the study contributes to the growing literature emphasizing the acute sensorimotor consequences of sedentary postures on balance control.

Several limitations should be considered when interpreting these findings. First, the relatively small sample size may limit the generalizability of the results and reduce statistical power for detecting subtle effects. Second, the use of convenience sampling from a single university population may introduce selection bias. Third, the study examined only acute responses to a single 30-minute slump sitting period and, therefore, cannot address long-term adaptations. Additionally, only female participants were included, limiting the applicability of findings to male populations. Furthermore, BMI was used as a proxy measure for body status; however, it does not account for body composition. Future studies should incorporate direct assessments such as body fat percentage or lean mass to provide a more accurate evaluation, particularly in athletic populations. The athlete group included participants from different sport disciplines and competitive levels, which may have influenced postural control outcomes. Finally, factors such as fatigue level and habitual posture were not controlled in detail and may have influenced balance outcomes. Incorporating electromyographic or kinematic analyses in future research may clarify the underlying mechanisms responsible for changes in postural sway.

Conclusion

In a nutshell, the present study demonstrated that prolonged slump sitting may influence postural sway in female athletes and non-athletes, with specific differences emerging between groups under certain sensory and stability conditions. By highlighting the potential impact of slump sitting on postural stability, this research contributes to the fields of sports science, ergonomics, and rehabilitation, emphasizing the need for strategies that mitigate the negative effects of prolonged sedentary behavior.

Abbreviations

LOS	Limits of Stability
mCTSIB	Modified Clinical Test of Sensory Interaction on Balance
PST	Postural Stability Test

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Not applicable.

Authors' contributions

RZ, HM, and MK contributed to the study design and data collection. HM and EE drafted the manuscript and made critical revisions. All authors read and approved the final manuscript.

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Data availability

The experimental data that support the findings of this study are available in Zenodo with the identifier: 10.5281/zenodo.18667751.

Declarations

Ethics approval and consent to participate

Based on the Declaration of Helsinki, all stages of the study were explained to the participants, and then written informed consent was obtained. Before starting the investigation, study approval was obtained from the Biomedical Research Ethics Committee of Tehran University (Ethics code: IR.UT.SPORT.REC.1403.08).

Consent for publication

We hereby confirm that all participants provided explicit written informed consent for the publication of their personal or clinical data, as well as any potentially identifiable images.

Competing interests

The authors declare no competing interests.

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